GROCERY SUBSTITUTION WORKBENCH

# FreshMatch

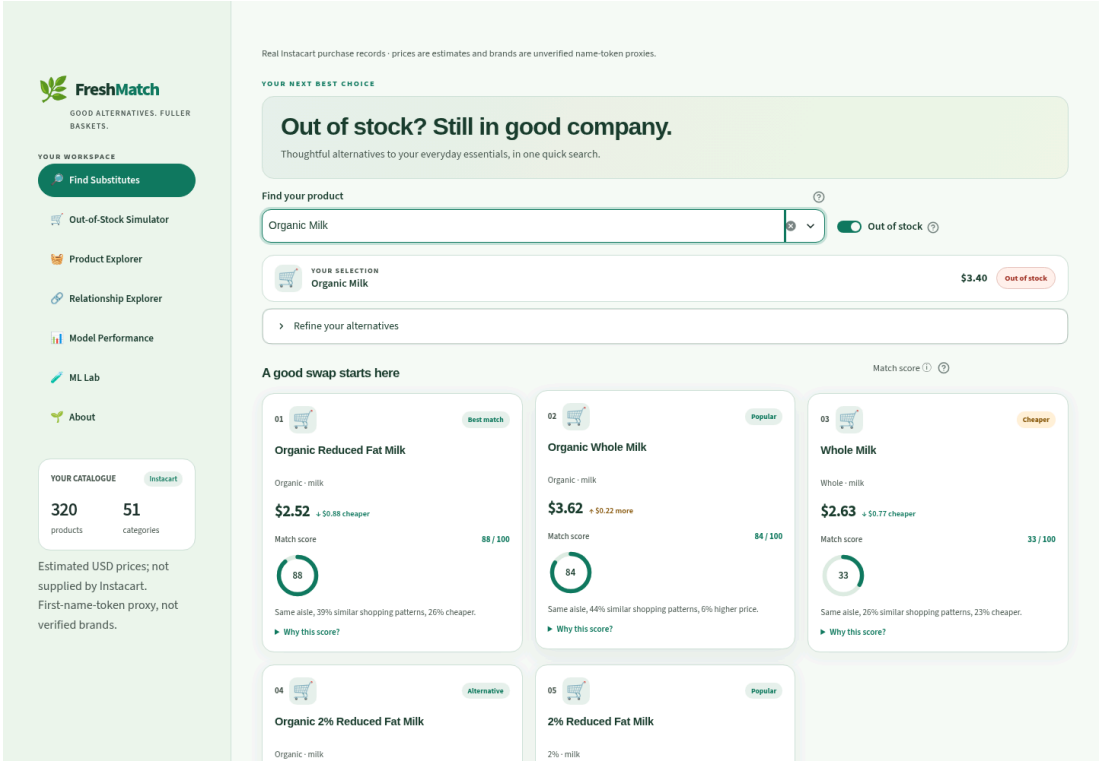
## Explainable grocery swaps using machine learning

Real purchase records. Clear comparisons.
A working app you can inspect.


PROJECT TEAM

K. Darshan Sai · RA2311027010002
J. Ranga Mokshagna · RA2311027010003
P. Bharat Sai · RA2311027010007


THE REAL-DATA APP




Python · XGBoost · LambdaMART · Streamlit

# A missing item should not end the basket

01

## A shopper needs milk

The preferred product is unavailable.

→

02

## Several choices exist

Names and prices alone are not enough.

→

03

## A useful swap is needed

Show relevant, available alternatives.

**Our goal**

**Find plausible alternatives, explain the match, and let the user control price and availability.**

# Real Instacart records, with a clear study sample

**49,688**
products in the source

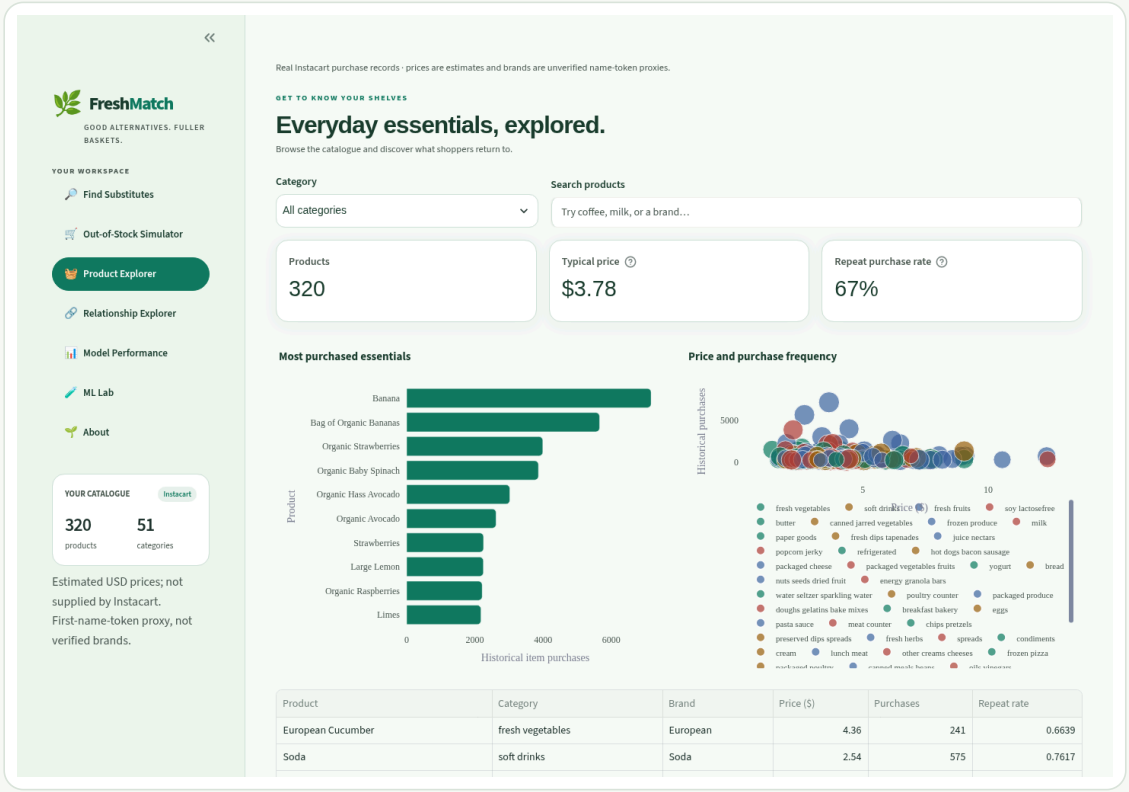
**320**
products in this study

**3,000**
sampled households

**49,760**
historical baskets

3,000 final baskets are held out.
The selected catalogue spans 51 aisles.

Real purchases; estimated prices and brand proxies.




Product Explorer · current real-data catalogue

# From basket history to an explained recommendation

01

## Validate records

CSV schemas and source hashes

→

02

## Learn relationships

Names, shopping context and co-purchase

→

03

## Train and compare

Six classifiers and a behavioral ranker

→

04

## Apply user filters

Same aisle, price cap and unavailable items

→

05

## Explain the result

Ranked alternatives with clear reasons

## Signals used by the models

### Product information

Category and TF-IDF name similarity

### Shopping behavior

Context, lift and reorder patterns

### Relative fit

Estimated price and popularity ratios

# Two learning tasks: suitability and behavior

SUITABILITY

## Six models, the same evaluation rules

Random Forest · Gradient Boosting

Logistic Regression · XGBoost

MLP · Soft voting

**Main app: calibrated XGBoost**

Selected by training-CV average precision.
Scores follow rule-based training labels.

BEHAVIORAL RANKING

## LambdaMART

Learns which alternatives appear in later purchase records.

8,840 training groups

147,340 candidate rows

Selected on validation NDCG@5:
depth 2, 160 trees.

# Train, choose and test on separate groups

## Suitability models: keep products separate

### 2,036 train pairs

Training with three validation folds

### 266 test pairs

Train and test use different products

### 1,532 pairs removed

Pairs crossing the split are removed

## Behavioral ranker: keep households separate

### 2,100 train households

Historical switches fit the ranker

### 450 validation households

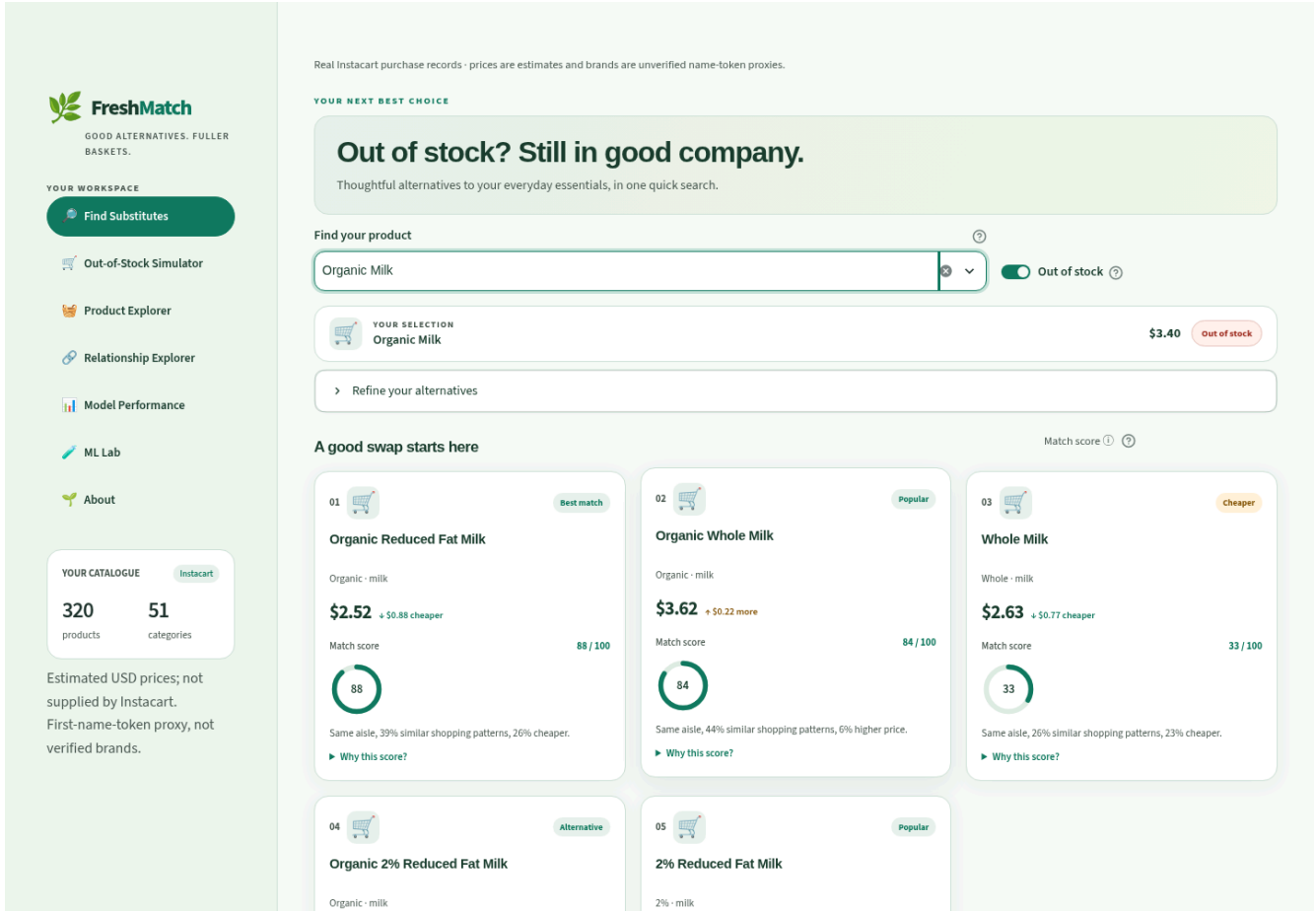
384 events choose the configuration

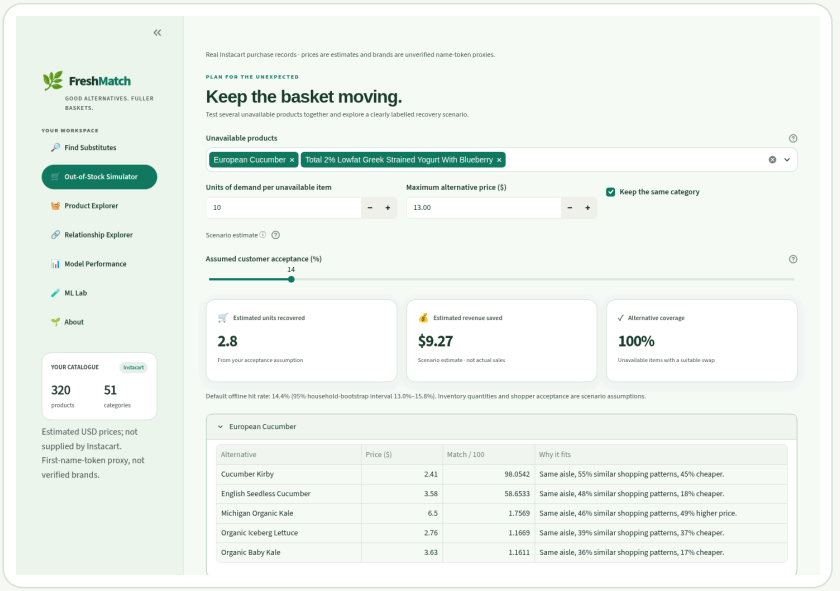
### 450 test households

365 events from 163 households

Limits: only 5 test pairs labelled as good swaps; history features still use all households.

# Find a swap, then test a stockout scenario



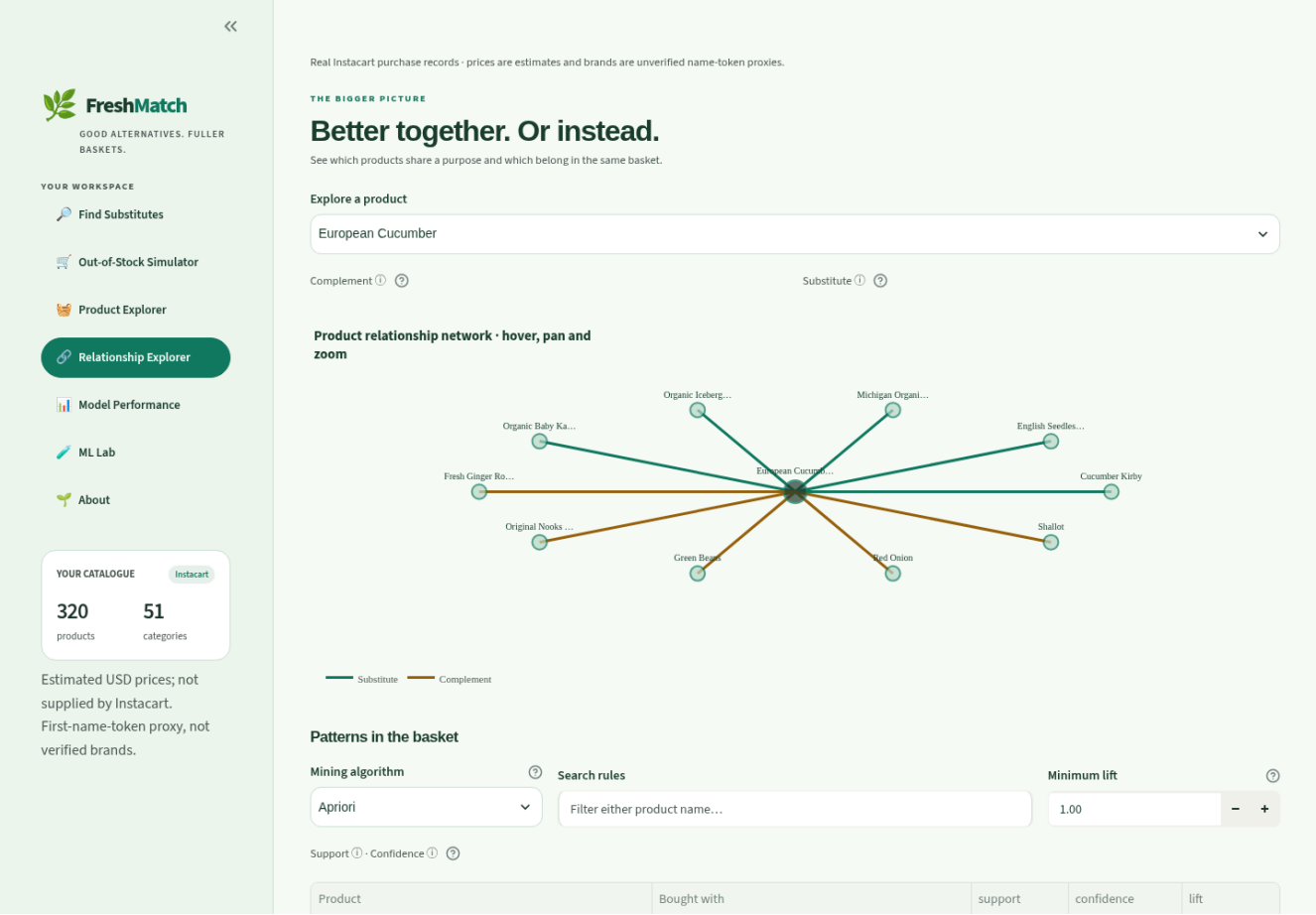

## Find Substitutes

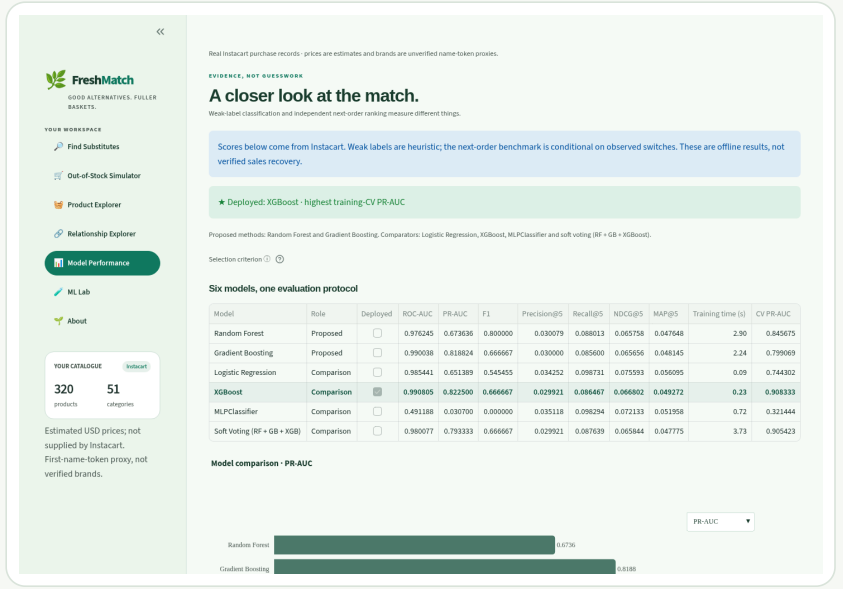
Search, filter and read the reason for each suggested alternative.

## Stockout Simulator

Explore an assumed recovery scenario.

# Show the relationship—and the evidence behind it




## Different relationships

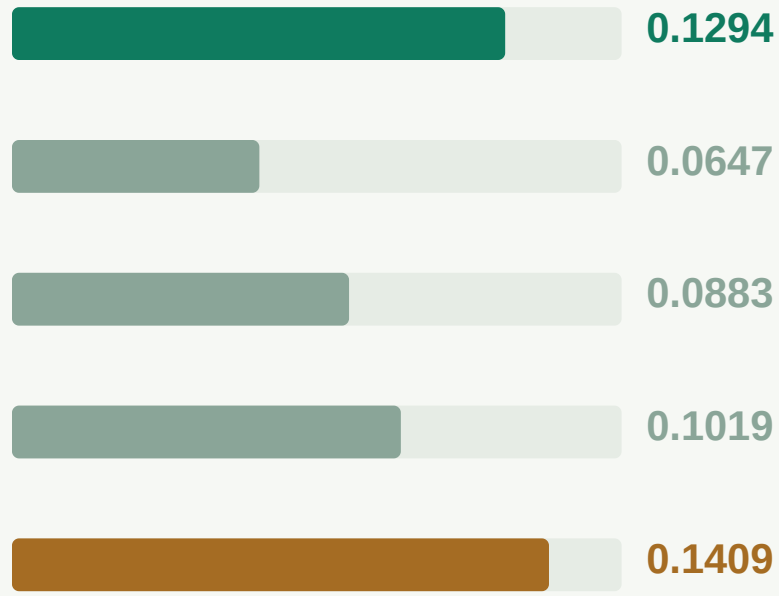
Complements are bought together.
Substitutes are possible replacements.

## Model Performance

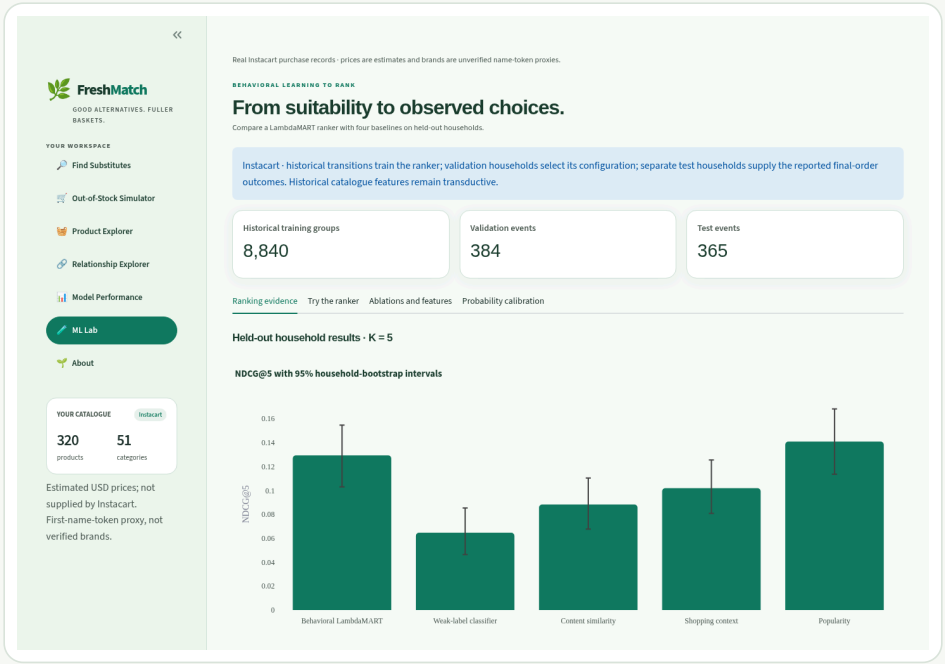
Compare results, errors and feature influence.

# The ranker improves on suitability; popularity still leads

## Ranking quality (NDCG@5) · higher is better




Same 365 test events · observed purchase switches




### 36.7% target coverage

Better candidate coverage is the next priority.

# A working research project, with clear next steps

## 23 tests passed

Seven working pages · reproducible real-data run

- **What works**

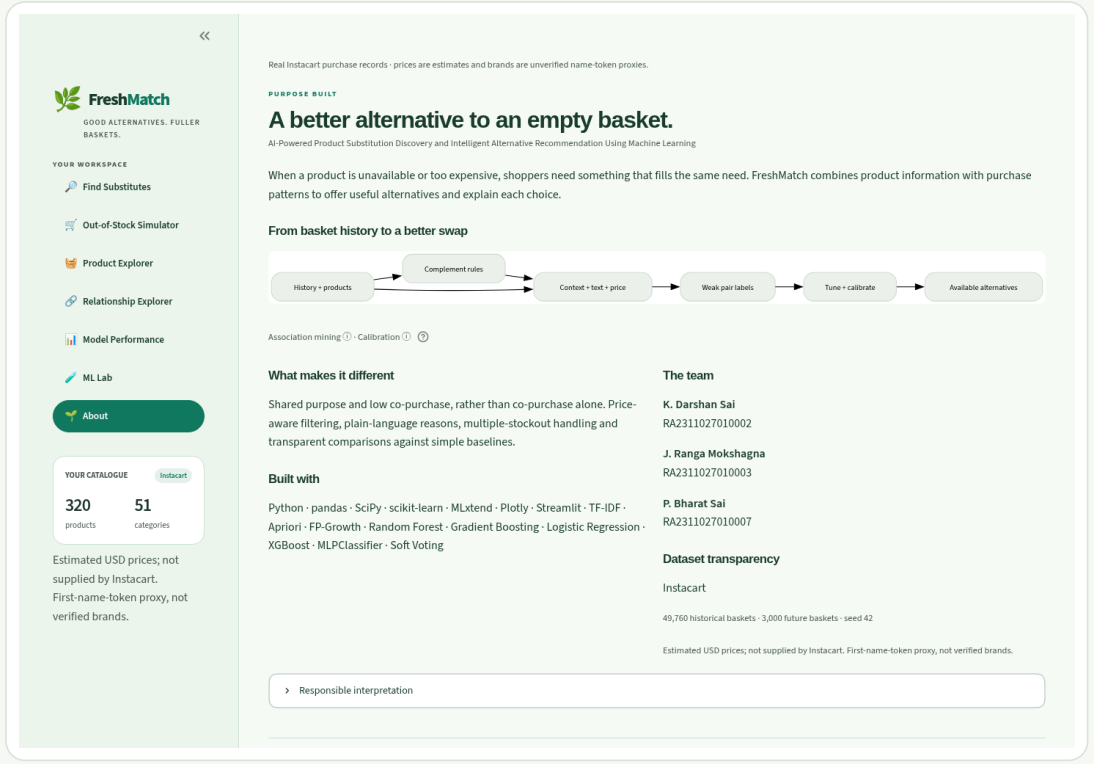
  Real purchase data, explained suggestions and transparent evaluation.

- **What comes next**

  Wider candidate coverage, expert labels and real price / ingredient data.

- **Limits to remember**

  Prices are estimates; dietary safety and actual sales recovery are unverified.




## Thank you. Questions?